



## Outline

## Covariates

## Covariates

## Transition-specific covariates

## Covariate modeling

## Proportional baseline hazards

## The Markov assumption

## Discussion







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## Transition-specific covariates

## Incorporated with proportional hazards

The transition hazard of transition  $k \rightarrow \ell$  for an individual with covariate vector  $\mathbf{Z}$  is given by

$$h_{kl}(t) = h_{kl,0}(t) \exp(\beta_{kl}^\top \mathbf{Z}),$$

where  $h_{k\ell,0}(t)$  is the baseline hazard and  $\beta_{k\ell}$  is the vector of covariate effects for transition  $k \rightarrow \ell$

## Transition-specific covariates

## Incorporated with proportional hazards

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## Transition-specific covariates

Equivalently, define **transition-specific covariates**  $\mathbf{z}_{k\ell}$  and

$$h_{k\ell}(t) = h_{k\ell,0}(t) \exp(\beta^\top \mathbf{Z}_{k\ell}),$$

where  $\beta$  is now a single vector containing the effects of the transition-specific covariates

## Transition-specific covariates

- ▶ Example: competing risks with two causes of failure

## Original data (wide format) ...

|   | id | time | status | Z |
|---|----|------|--------|---|
| 1 | 1  | 6    | 0      | 1 |
| 2 | 2  | 4    | 1      | 3 |
| 3 | 3  | 5    | 2      | 6 |

## Transition-specific covariates

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## Original data (wide format) ...

|   | id | time | status | Z |
|---|----|------|--------|---|
| 1 | 1  | 6    | 0      | 1 |
| 2 | 2  | 4    | 1      | 3 |
| 3 | 3  | 5    | 2      | 6 |

... in long (mstate) format with cause-specific covariates:

|   | id | from | to | trans | Tstart | Tstop | time | status | Z.1 | Z.2 |
|---|----|------|----|-------|--------|-------|------|--------|-----|-----|
| 1 | 1  | 1    | 2  | 1     | 0      | 6     | 6    | 0      | 1   | 0   |
| 2 | 1  | 1    | 3  | 2     | 0      | 6     | 6    | 0      | 0   | 1   |
| 3 | 2  | 1    | 2  | 1     | 0      | 4     | 4    | 1      | 3   | 0   |
| 4 | 2  | 1    | 3  | 2     | 0      | 4     | 4    | 0      | 0   | 3   |
| 5 | 3  | 1    | 2  | 1     | 0      | 5     | 5    | 0      | 6   | 0   |
| 6 | 3  | 1    | 3  | 2     | 0      | 5     | 5    | 1      | 0   | 6   |





## Why does it work?

- ▶ Model:  $h_k(t | Z) = h_{k0}(t) \exp(\beta_k Z)$ ,  $k = 1, 2$
- ▶ Likelihood contribution:

$$(h_{10}(t_j) \exp(\beta_1 Z_j))^{\mathbf{1}\{d_j=1\}} (h_{20}(t_j) \exp(\beta_2 Z_j))^{\mathbf{1}\{d_j=2\}} \\ \exp(-(H_{10}(t_j) \exp(\beta_1 Z_j) + H_{20}(t_j) \exp(\beta_2 Z_j)))$$

- ▶ With cause-specific covariates, model (stratified by cause/transition):

$$h(t | Z_1, Z_2, \text{cause } k) = h_{k0}(t) \exp(\beta_1 Z_1 + \beta_2 Z_2)$$

- ▶ Likelihood contribution of subject  $i$  is product of two data lines

## 1. First line

$$(h_{10}(t_i) \exp(\beta_1 Z_i + \beta_2 0))^{\mathbf{1}\{d_i=1\}} \exp(-(H_{10}(t_i) \exp(\beta_1 Z_i + \beta_2 0)))$$

## 2. Second line

$$(h_{20}(t_i) \exp(\beta_1 0 + \beta_2 Z_i))^{1\{d_i=2\}} \exp(-(H_{20}(t_i) \exp(\beta_1 0 + \beta_2 Z_i)))$$



## Covariates in CLL data ( $n = 724$ )

| Prognostic factor |                            | $n$ (%)          |
|-------------------|----------------------------|------------------|
| Donor             | HLA-id sib donor           | 291 (40)         |
|                   | Other                      | 433 (60)         |
| Remission status  | Complete Remission         | 94 (13)          |
|                   | Partial Remission          | 366 (51)         |
|                   | Stable/progressive disease | 264 (36)         |
| Sex match         | Other                      | 567 (78)         |
|                   | Female to male             | 157 (22)         |
| Age at SCT        |                            | 55 (19-74)       |
| Year of SCT       |                            | 2007 (2000-2011) |

► Age centered and rescaled:  $\text{age10} = (\text{age}-55)/10$

# Modeling transitions separately

## Transition 1

```
> cox1 <- coxph(Surv(Tstart, Tstop, status) ~ age10 + remstat + SIBdonor +
+               data=msc11, subset=trans==1, method="breslow")
> cox1
```

Call:

```
coxph(formula = Surv(Tstart, Tstop, status) ~ age10 + remstat +
      SIBdonor + sexmatch, data = msc11, subset = trans == 1, method = "breslow")
```

|                        | coef    | exp(coef) | se(coef) | z      | p       |
|------------------------|---------|-----------|----------|--------|---------|
| age10                  | -0.0172 | 0.983     | 0.0957   | -0.179 | 0.86000 |
| remstatPR              | 1.0932  | 2.984     | 0.3331   | 3.282  | 0.00100 |
| remstatSD/PD           | 1.2683  | 3.555     | 0.3376   | 3.757  | 0.00017 |
| SIBdonorother donor    | -0.0708 | 0.932     | 0.1539   | -0.460 | 0.65000 |
| sexmatchfemale to male | 0.3584  | 1.431     | 0.1754   | 2.043  | 0.04100 |

Likelihood ratio test=23 on 5 df, p=0.000343 n= 724, number of events= 10



# Modeling transitions separately

## Transition 2

```
> cox2 <- coxph(Surv(Tstart, Tstop, status) ~ age10 + remstat + SIBdonor
+ data=msc11, subset=trans==2, method="breslow")
> cox2
```

Call:

```
coxph(formula = Surv(Tstart, Tstop, status) ~ age10 + remstat +
      SIBdonor + sexmatch, data = msc11, subset = trans == 2, method = "breslow")
```

|                        | coef   | exp(coef) | se(coef) | z      | p       |
|------------------------|--------|-----------|----------|--------|---------|
| age10                  | 0.3720 | 1.45      | 0.0866   | 4.2971 | 1.7e-05 |
| remstatPR              | 0.0164 | 1.02      | 0.2098   | 0.0783 | 9.4e-01 |
| remstatSD/PD           | 0.4959 | 1.64      | 0.2104   | 2.3569 | 1.8e-02 |
| SIBdonorother donor    | 0.4850 | 1.62      | 0.1414   | 3.4303 | 6.0e-04 |
| sexmatchfemale to male | 0.4190 | 1.52      | 0.1520   | 2.7577 | 5.8e-03 |

Likelihood ratio test=50.9 on 5 df, p=9.22e-10 n= 724, number of events= 10



# Modeling transitions separately

## Transition 3

```
> cox3 <- coxph(Surv(Tstart, Tstop, status) ~ age10 + remstat + SIBdonor +
+               data=msc11, subset=trans==3, method="breslow")
> cox3
```

Call:

```
coxph(formula = Surv(Tstart, Tstop, status) ~ age10 + remstat +
      SIBdonor + sexmatch, data = msc11, subset = trans == 3, method = "breslow")
```

|                        | coef    | exp(coef) | se(coef) | z      | p    |
|------------------------|---------|-----------|----------|--------|------|
| age10                  | 0.0242  | 1.025     | 0.118    | 0.205  | 0.84 |
| remstatPR              | -0.1537 | 0.858     | 0.439    | -0.350 | 0.73 |
| remstatSD/PD           | -0.1088 | 0.897     | 0.441    | -0.247 | 0.81 |
| SIBdonorother donor    | -0.2139 | 0.807     | 0.198    | -1.079 | 0.28 |
| sexmatchfemale to male | 0.2023  | 1.224     | 0.230    | 0.879  | 0.38 |

Likelihood ratio test=2.51 on 5 df, p=0.775 n= 170, number of events= 5



# Modeling transitions together

- ▶ We can also model all transitions in a single Cox regression model, using stratified regression and transition-specific covariates
- ▶ First call `expand.covs` to define them

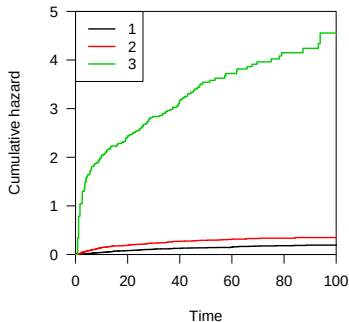
```
> covs <- c("age10", "remstat", "SIBdonor", "sexmatch")
> msc11 <- expand.covs(msc11, covs, longnames=FALSE)
> cox0 <- coxph(Surv(Tstart, Tstop, status) ~
+   age10.1 + remstat1.1 + remstat2.1 + SIBdonor.1 + sexmatch.1
+   age10.2 + remstat1.2 + remstat2.2 + SIBdonor.2 + sexmatch.2
+   age10.3 + remstat1.3 + remstat2.3 + SIBdonor.3 + sexmatch.3
+   + strata(trans), data=msc11, method="breslow")
> cox0
```

|            | coef    | exp(coef) | se(coef) | z       | p       |
|------------|---------|-----------|----------|---------|---------|
| age10.1    | -0.0172 | 0.983     | 0.0957   | -0.1794 | 8.6e-01 |
| remstat1.1 | 1.0932  | 2.984     | 0.3331   | 3.2818  | 1.0e-03 |
| remstat2.1 | 1.2683  | 3.555     | 0.3376   | 3.7569  | 1.7e-04 |
| SIBdonor.1 | -0.0708 | 0.932     | 0.1539   | -0.4604 | 6.5e-01 |
| sexmatch.1 | 0.3584  | 1.431     | 0.1754   | 2.0425  | 4.1e-02 |
| age10.2    | 0.3720  | 1.451     | 0.0866   | 4.2971  | 1.7e-05 |
| remstat1.2 | 0.0164  | 1.017     | 0.2098   | 0.0783  | 9.4e-01 |
| remstat2.2 | 0.4959  | 1.642     | 0.2104   | 2.3569  | 1.8e-02 |
| SIBdonor.2 | 0.4850  | 1.624     | 0.1414   | 3.4303  | 6.0e-04 |
| sexmatch.2 | 0.4190  | 1.521     | 0.1520   | 2.7577  | 5.8e-03 |
| age10.3    | 0.0242  | 1.025     | 0.1181   | 0.2051  | 8.4e-01 |
| remstat1.3 | -0.1537 | 0.858     | 0.4393   | -0.3498 | 7.3e-01 |
| remstat2.3 | -0.1088 | 0.897     | 0.4411   | -0.2467 | 8.1e-01 |
| SIBdonor.3 | -0.2139 | 0.807     | 0.1983   | -1.0786 | 2.8e-01 |
| sexmatch.3 | 0.2023  | 1.224     | 0.2302   | 0.8787  | 3.8e-01 |

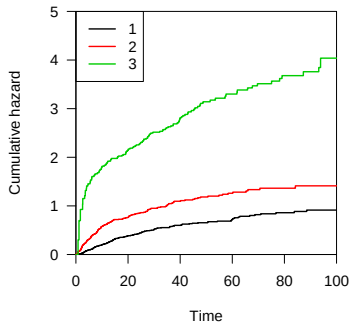
Likelihood ratio test=76.3 on 15 df, p=3.25e-10 n= 1618, number of e

# Cumulative hazards (semi-parametric model)

Good risk patient, trans-spec model



Poor risk patient, trans-spec model



# Advantages of this approach

- ▶ Why so complicated?
- ▶ Reason: tremendous flexibility in
  - ▶ varying/restricting covariate effects across transitions
  - ▶ combining baseline hazards of different transitions

## Varying/restricting covariate effects

- ▶ Using transition-specific covariates, you obtain different covariate effect estimates across transitions
- ▶ Using the global covariate itself, you obtain (stratified Cox regression) a single effect estimate assumed to be common for all transitions
- ▶ Somewhere in between is also possible (stratum-specific covariates)
- ▶ The effect of sex mismatch could be assumed to be identical for transitions 2 and 3, and different for transition 1
  - ▶ Use `sexmatch.1` along with new covariate `sexmatch.23 = sexmatch.2 + sexmatch.3`
- ▶ Any combination of such common and transition-specific covariates is also possible
- ▶ You can also leave out covariates for one transition and include them for another transition

## For example

```
> coxph(Surv(Tstart, Tstop, status) ~ age10.1 + age10.2 + age10.3  
      + SIBdonor + sexmatch.1 + sexmatch.23 + remstat.3 + strata(trans),  
      data=msc11, method="breslow")
```

would fit a Markov model with

- ▶ Three age effects, one for each transition
- ▶ One common donor effect for all transitions
- ▶ Two sex mismatch effects, one for transition 1, one (common) for transitions 2 and 3
- ▶ One remission status effect for transition 3 only (assumed to be zero for transitions 1 and 2)
- ▶ Three baseline hazards, one for each transition (`strata(trans)`)

## Proportional baseline hazards

- ▶ **Proportional hazards assumption**, hazard rates going into the same state are often assumed to be proportional
- ▶ Here, the two transitions into Death are assumed proportional:
  - ▶ SCT  $\rightarrow$  NRM (transition 2,  $1 \rightarrow 3$ )
  - ▶ Rel  $\rightarrow$  Death after Rel (transition 3,  $2 \rightarrow 4$ )
- ▶ Modeled with time-dependent covariate, indicating whether or not intermediate event (Rel) has occurred

$$Z_{Rel}(t) = \begin{cases} 0, & \text{if } t \leq \text{time of Rel} \\ 1, & \text{if } t > \text{time of Rel} \end{cases}$$

- ▶ Hazard rates for  $1 \rightarrow 3$  and  $2 \rightarrow 3$  transitions are given by

$$h(t) = h_0(t) \exp(\gamma Z_{Rel}(t)) ,$$

- ▶  $Z_{Rel}(t) = 0 \Leftrightarrow h(t) = h_{13}(t)$ , baseline hazard rate is  $h_0(t)$
- ▶  $Z_{Rel}(t) = 1 \Leftrightarrow h(t) = h_{23}(t)$ , baseline hazard =  $h_0(t) \exp(\gamma)$

## Proportional baseline hazards (ctd)

- ▶ Two advantages to this approach
  - ▶ Have to estimate one hazard (plus a coefficient) instead of two (minor advantage)
  - ▶  $\exp(\gamma_{Rel})$  is hazard ratio of occurrence of intermediate state Rel; indicates how much occurrence of Rel influences risk of death (major advantage)
- ▶ Disadvantage: it is an assumption, does it hold?

NB: Note the similarity with time-dependent Cox models - but the multi-state model includes modelling occurrence of intermediate event

## Proportional baseline hazards

```
coxPBH <- coxph(Surv(Tstart, Tstop, status) ~
  age10.1 + remstat1.1 + remstat2.1 + SIBdonor.1 + sexmatch.1 +
  age10.2 + remstat1.2 + remstat2.2 + SIBdonor.2 + sexmatch.2 +
  age10.3 + remstat1.3 + remstat2.3 + SIBdonor.3 + sexmatch.3 +
  + strata(strata), data=msc11, method="breslow")
```

|            | coef     | exp(coef) | se(coef) | z      | Pr(> z ) |     |
|------------|----------|-----------|----------|--------|----------|-----|
| age10.1    | -0.01717 | 0.98297   | 0.09573  | -0.179 | 0.857624 |     |
| remstat1.1 | 1.09321  | 2.98384   | 0.33311  | 3.282  | 0.001032 | **  |
| remstat2.1 | 1.26831  | 3.55483   | 0.33760  | 3.757  | 0.000172 | *** |
| SIBdonor.1 | -0.07083 | 0.93162   | 0.15386  | -0.460 | 0.645247 |     |
| sexmatch.1 | 0.35835  | 1.43097   | 0.17545  | 2.043  | 0.041099 | *   |
| age10.2    | 0.37672  | 1.45749   | 0.08648  | 4.356  | 1.32e-05 | *** |
| remstat1.2 | 0.01998  | 1.02018   | 0.20982  | 0.095  | 0.924126 |     |
| remstat2.2 | 0.49102  | 1.63398   | 0.21029  | 2.335  | 0.019545 | *   |
| SIBdonor.2 | 0.49836  | 1.64603   | 0.14104  | 3.534  | 0.000410 | *** |
| sexmatch.2 | 0.42081  | 1.52319   | 0.15186  | 2.771  | 0.005587 | **  |
| age10.3    | 0.01832  | 1.01849   | 0.11629  | 0.158  | 0.874820 |     |
| remstat1.3 | -0.11961 | 0.88727   | 0.43395  | -0.276 | 0.782840 |     |
| remstat2.3 | -0.14905 | 0.86153   | 0.43697  | -0.341 | 0.733033 |     |
| SIBdonor.3 | -0.20624 | 0.81364   | 0.19518  | -1.057 | 0.290662 |     |
| sexmatch.3 | 0.20641  | 1.22925   | 0.22645  | 0.911  | 0.362041 |     |
| Z2.3       | 2.25895  | 9.57300   | 0.48390  | 4.668  | 3.04e-06 | *** |

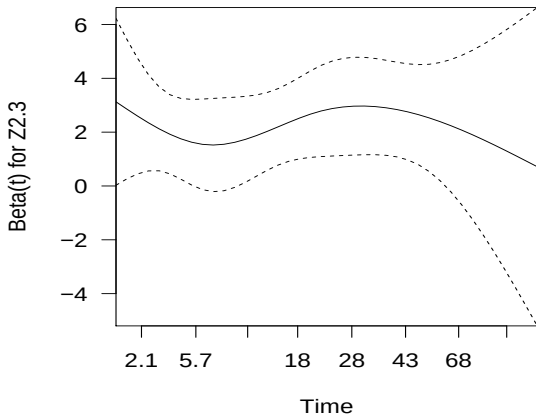
## Results II

- ▶ Relapse has a strongly significant detrimental effect (HR=9.6, 95% CI: 3.7 – 24.7)
- ▶ This holds for two patients that have the same values of the other covariates
- ▶ Proportional hazards assumption may be tested

```
> cox.zph(coxPBH)
```

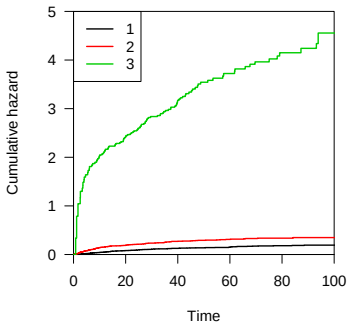
|      | rho      | chisq    | p      |
|------|----------|----------|--------|
| z2.3 | 0.010265 | 5.53e-02 | 0.8140 |

# Model fit

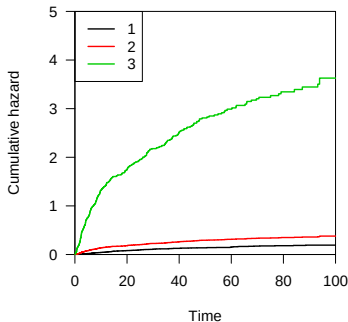


## Comparison of stratified and proportional hazards

**Good risk patient, trans-spec model**



### Good risk patient, PBH model





## Which model to be preferred?

- ▶ In case of few events/many covariates, stratifying/combining information is necessary
- ▶ Hazard ratio for relapse is useful here, but proportionality assumption needs to be checked in general
- ▶ Consider pre-selection of covariates on clinical grounds
- ▶ Possibly, use further statistical selection procedures

# Flexibility

- ▶ IF HR intermediate event is close to 1, we could decide to omit  $z_2$  as covariate;  $\Rightarrow$  fit a model where we assume both baseline transition hazards into Death to be equal - but note interaction with other covariates
- ▶ We could in principle also assume all transitions to be proportional; a priori this assumption is not likely to hold, however
- ▶ In summary: there is tremendous flexibility in modeling the effect of covariates using long data format, stratified regression, and time-dependent and transition-specific covariates

## Testing the Markov assumption

- ▶ The Markov assumption can be tested by adding the time of entering the Relapse state for the transition Rel  $\rightarrow$  Death
- ▶ Little trick: add time of relapse as a "covariate" to the data, and later use `expand.covs` to define transition-specific versions
- ▶ Simply add `reltime.3` to the model
- ▶ Significance of `reltime.3` in the model would be evidence of violation of the Markov assumption
- ▶ A p-value  $> 0.05$  does not guarantee that the Markov assumption is true (other than type II error); there may be other reasons why the Markov assumption doesn't hold
- ▶ What to do if the Markov assumption is violated?
  - ▶ Non-parametric methods (de Una-Alvarez)
  - ▶ Semi-Markov (clock reset) models

## Clock reset approach

- ▶ Another reason why the Markov assumption doesn't hold could be duration dependence
- ▶ The transition hazard of Rel  $\rightarrow$  Death could be a function of time since entrance of the Rel state
- ▶ The clock is reset as soon as a new state is entered
- ▶ Perhaps biologically more plausible in some situations
- ▶ For instance if the intermediate state has early new events (transplantation for dialysis patients, which is associated with early transplant-related mortality)
- ▶ If Markov assumption holds within this transition, then the resulting process is called a Markov renewal process, the model is called a (homogeneous) semi-Markov model (clock reset approach)

## Estimation in clock reset models

- ▶ Not really much different from what we have seen for Markov models
- ▶ The techniques shown before carry through
  - ▶ Long format, stratified regression, transition-specific covariates
  - ▶ Assumption of proportional baseline transition hazards less natural than in clock forward models
  - ▶ No delayed entry, use `Surv(time, status)` in R
- ▶ Results similar to those obtained in the Markov model (not shown)

## Patient-specific transition intensities

- ▶ This will become crucial for prediction
- ▶ Given a model, and associated estimates of (expanded) covariate effects  $\hat{\beta}$  and baseline hazards  $\Delta\hat{H}_{q0}(t)$  for all strata  $q = 1, \dots, Q$
- ▶ What are the estimated transition intensities corresponding to a patient with basic covariate vector  $\mathbf{Z}_0$ ?
- ▶ For transition  $g \rightarrow h$ , denote the corresponding expanded transition-specific covariate by  $\mathbf{Z}^*$ , and the stratum by  $q$
- ▶ Patient-specific transition intensity at  $t$  given by

$$\exp(\hat{\beta}^\top \mathbf{Z}^*) \Delta\hat{H}_{q0}(t)$$

- ▶ Variances and covariances (for different transitions) may be obtained; implemented in `mstate`

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## Dynamic prediction

- ▶ Want to answer questions like:
  - ▶ Given a Bone Marrow Transplant patient who has had acute Graft-versus-Host Disease after 60 days, and whose platelets have recovered after 120 days and who has had no further events after 1 year, what is then the probability of surviving relapse-free for 2 more years?
  - ▶ Given a breast cancer patient who is alive without local recurrence and distant metastasis at 1.5 years after surgery, what are the probabilities of remaining alive and event-free, of experiencing a local recurrence and/or distant metastasis, etc, for the coming 10 years?

# Summary

- ▶ Non-parametric estimation of transition hazards is straightforward
- ▶ Great flexibility in modeling the effect of covariates using long data format, stratified regression, and time-dependent and transition-specific covariates
- ▶ Note: as in competing risks, covariate effects on transition intensities do not have a 1-1 relation with effects on transition probabilities

## References



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Multi-state models for event history analysis.  
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